NETWORK MONITORING AND TROUBLESHOOTING ASSIGNMENT

1. Install Wireshark and capture 2 minutes of network traffic while browsing Instagram or YouTube. Save the capture as a .pcap file and note the top 3 protocols.

Procedure

- Install Wireshark from its official website and open it.
- Select the active Wi-Fi or Ethernet network interface.
- Start capturing, browse Instagram or YouTube for approximately 2 minutes, and then stop the capture.
- Choose File > Save As and save the capture with a .pcap or .pcapng extension.

Finding the top protocols: Open Statistics > Protocol Hierarchy in Wireshark. Note the three protocols with the largest packet counts or percentages.

Protocol	Packets / percentage observed
1. Frame	43
2. Ethernet	43
3. IPv4	41

Capture file name: test.pcapng

2. Use ping to test connectivity to www.spotify.com and www.zomato.com. Record the response times and identify which site responds faster.

ping www.spotify.com
ping www.zomato.com

Website	Average response time	Packet loss
www.spotify.com	24.589 ms	0 %
www.zomato.com	16.442 ms	0 %

Faster website: Zomato

Explanation: A lower average response time means the replies returned more quickly from that test. Results can change depending on network conditions, server location, and routing.

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list running processes related to Chrome, WhatsApp Desktop, or another browser-based app. Record the process name and memory usage.

Procedure

Windows: Open Task Manager with Ctrl + Shift + Esc. Select the Processes tab and check the Memory column. Expand app groups if needed to see individual processes.

Application	Process name	Memory usage
Chrome / browser	Firefox	451 MB
WhatsApp Desktop / other app	Whatsapp	303 MB
Other browser-based app	Vivaldi browser	301 MB

4. Download and run Angry IP Scanner or Advanced IP Scanner to scan your local Wi-Fi network. List the IP address and device name of at least 2 devices.

Procedure

- Download one of the scanners from its official website.
- Find your local network range. For example, an address such as 192.168.1.9 with a /24 network commonly belongs to the range 192.168.1.1–192.168.1.254.
- Enter only your own home or authorized college network range and start the scan.
- Record at least two discovered devices and their IP addresses and hostnames, if available.

Device number	IP address	Device name / hostname
1	192.168.1.3	192.168.1.3
2	192.168.1.9	192.168.1.9

5. Suggest 3 hardware tools used in network troubleshooting and describe when each would be used.

Hardware tool	When it is used
1. Cable tester	Used to check Ethernet cables for broken wires, incorrect wiring, or faulty connections.
2. Crimping tool	Used to attach RJ45 connectors to Ethernet cables when making or repairing network cables.
3. Multimeter	Used to measure voltage, resistance, and continuity when checking power supplies, cables, or electrical connections.

Conclusion: Network troubleshooting involves checking traffic, connectivity, running applications, and connected devices. Software tools help identify network problems, while hardware tools help test and repair physical connections.